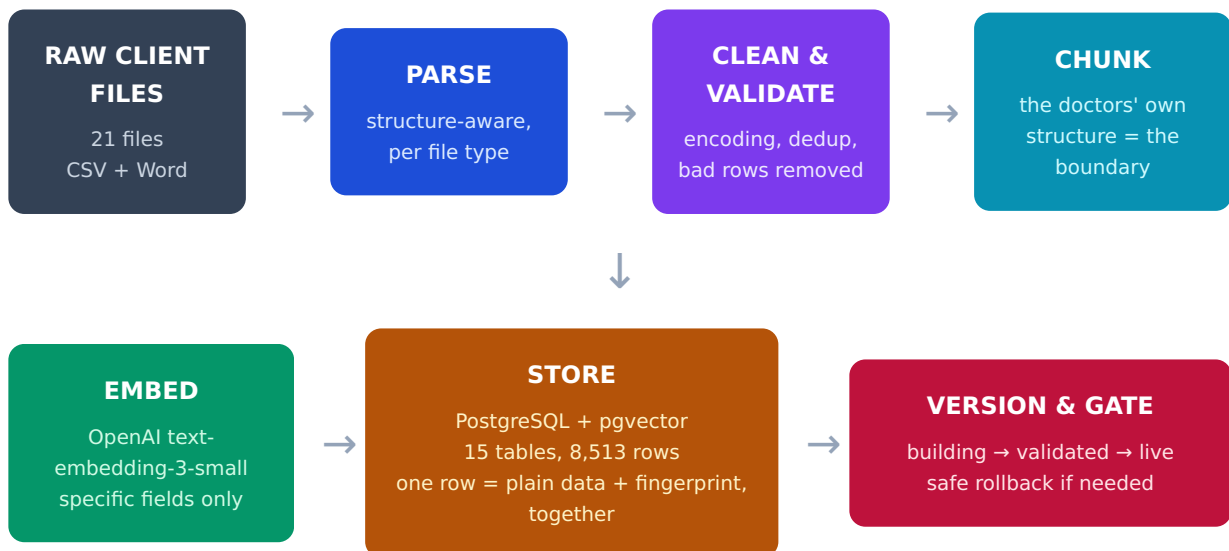


26. Ingestion Pipeline Architecture — How the Client's Data Became a Searchable Knowledge Base

Purpose

This document explains how every doctor-written file the client provided was turned into a structured, searchable database — the foundation everything else in this project is built on. It covers **ingestion only** (raw files → database), not how the AI later retrieves and uses that data in a live conversation, which is documented separately.

1. The architecture, end to end



This runs **offline, once per data update** — never while a real user is waiting for a reply. Speed here doesn't matter; correctness does.

2. What happens at each stage

Stage 1 — Raw client files. 21 distinct files across two drops: 6 original clinical spreadsheets (signals, symptoms, red flags, routing rules, severity thresholds, fallback logic) and 15 files from a September 2026 drop (layer/depth question banks, emotion phrasing, scenario and career data, self-help material). Formats: CSV, Excel-derived CSV exports, and Word documents — some password-protected, one built entirely from image-style text boxes rather than normal paragraphs.

Stage 2 — Structure-aware parsing. Each file gets its own dedicated parser, because each one is laid out differently — some are clean spreadsheets, some mix tables and headings in a single Word document, one is built from Microsoft Word's drawing-canvas feature instead of plain text. A shared parsing toolkit handles the common problems (Windows-1252 encoding instead of UTF-8, forward-filling labels that were only written once per group, walking documents in their real reading order rather than paragraphs-then-tables).

Stage 3 — Cleaning & validation. Real spreadsheet exports carry real mess: instructional template rows left behind by the client's form, duplicated content from Word's internal rendering quirks, missing values that could silently corrupt data if handled carelessly. Every file was checked against its own real structure — not assumed — before being trusted.

Stage 4 — Chunking. For every file received so far, the client's own structure already is the right chunk boundary — one spreadsheet row, or one paragraph under a heading, is already one complete, self-contained idea. No AI is used to decide how to split content; that's reserved for future prose-style content that isn't pre-segmented, which hasn't arrived yet.

Stage 5 — Embedding. Specific fields — never a whole record — are converted into a numeric "fingerprint" (an embedding) that captures meaning, not exact wording, using OpenAI's `text-embedding-3-small` model. This is what later lets the system recognize "I feel like I have nothing left to give" as the same idea as a differently-worded example in the database.

Stage 6 — Structured storage. Everything lands in one PostgreSQL database with the `pgvector` extension enabled — plain columns (IDs, categories, real text) and the numeric fingerprint sitting in the very same row, the same table. There is no separate "vector database" to keep in sync.

Stage 7 — Versioning & safe rollout. Every ingestion run creates a new, clearly labeled version (`building` → validated → `live`), so a knowledge-base update can be tested before it ever reaches a real conversation, and rolled back cleanly if something's wrong with it.

3. The real numbers

	Count
Source files ingested	21
Database tables populated	15
Total rows stored	8,513
Rows with a real vector embedding	8,294 (97%)

Table	Rows	What it holds
<code>kb_scenarios</code>	2,809	Real-life situation phrases
<code>kb_tier_question_bank</code>	1,413	Follow-up questions per scenario/career topic
<code>kb_layer_content</code>	1,512	Conversational-depth question/statement bank
<code>kb_emotion_phrases</code>	1,276	Real first-person emotion phrasing
<code>kb_career_lookup</code>	500	Career → field/stream reference
<code>kb_signals</code>	285	Clinical warning signs
<code>kb_sufficiency_patterns</code>	195	"How much evidence is enough" thresholds
<code>kb_symptoms</code>	154	Physical/emotional symptom descriptions
<code>kb_clarification_phrases</code>	115	"I'm not sure, let me ask" phrases
<code>kb_fallback_scenarios</code>	99	What to do when a conversation goes off-script
<code>kb_routing_rules</code>	66	Deterministic severity-routing rules
<code>kb_anxiety_situations</code>	35	Anxiety topic coverage checklist
<code>kb_red_flags</code>	31	Danger phrases + safety scripts
<code>kb_guardrail_rules</code>	18	Parenting-concern urgency rules
<code>kb_content_chunks</code>	5	Self-help tools (DBT skills)

4. A concrete, worked example

Before (one row spreadsheet cell):

flag_id: RF-001
trigger_phrases: "he overtook me on the road, so i followed him to beat him up"
| (4 more)
clinical_meaning: aggressive behaviour and irritability/violence risk
immediate_action: Tier 3 + SOS flag
escalation_target: Psychiatrist + crisis review

After ingestion (one database row, `kb_red_flags`):

Column	Value
flag_id	RF-001
trigger_phrases	the 5 phrases, as real text
trigger_embedding	[0.0231, -0.114, 0.442, ...] — 1024 numbers
clinical_meaning	"aggressive behaviour and irritability/violence risk"
escalation_target	"Psychiatrist + crisis review"

Nothing about the row's real content changed — the embedding is simply one more column sitting alongside it, in the same table, in the same database.

5. Quality and reliability measures taken

Every one of the 21 files was inspected directly against its own real structure before a parser was written for it — not assumed from a template. This surfaced and fixed several real, previously undiscovered issues along the way: hidden template-instruction rows in spreadsheet exports that looked like real data, a Word document built entirely from image-style drawing objects rather than normal text, duplicated content from a rendering quirk in one workbook, and column names containing a typographic dash character that a first-pass parser missed by one character — each confirmed against the live file and corrected before being trusted.

6. Honest scope — what's not in here yet

- **Phase III/IV self-care and psycho-education content** (beyond the 5 DBT skills already ingested) has not been received from the client yet — this is a content gap, not an ingestion limitation.

- **~20 style/tone example conversations** (Bucket 1 content) are deliberately not part of this pipeline — they're meant to teach the AI's tone directly, not be searched, and are handled separately.
- **3 files remain password-locked** with a different password than the rest of the September drop.
- **A small number of files** (a couple of self-help worksheets, one inconsistently-formatted conversation file) are reviewed and catalogued, but don't yet have a dedicated parser.